

Length-aware shaping. For prompt group rewards r_i , response lengths L_i , correct set $C = \{i : r_i > 0\}$, and $L_{\min} = \min_{j \in C} L_j$, incorrect responses keep raw reward.

Linear: $r'_i = r_i \max\left(0, 1 - \alpha \frac{L_i - L_{\min}}{L_{\min}}\right)$

Exponential: $r'_i = r_i \exp\left(-\lambda \frac{L_i - L_{\min}}{L_{\min}}\right)$

Ranked: $r'_i = \frac{r_i}{1 + \text{Rank}(L_i)}$

Binary shortest: $r'_i = r_i \mathbf{1}[L_i = L_{\min}]$

Soft threshold: $T = \text{median}\{L_j : j \in C\}, \quad r'_i = \begin{cases} r_i, & L_i \leq T \\ r_i \max\left(0, 1 - \frac{L_i - T}{T}\right), & L_i > T \end{cases}$

Warm-up.

$$r_i^{\text{train}} = (1 - w_t)r_i + w_t r'_i, \quad w_t = \min\left(1, \frac{\text{num_steps}}{\text{num_total_steps}}\right).$$

Alternatively, solve-rate warm-up keeps $w_t = 0$ until the batch solve rate first hits a chosen threshold; once it has hit, $w_t = 1$ for the rest of training.